Date:23.11.2020

2020-2021 FALL SEMESTER HYDRAULICS COURSE

Name: **Surname:** **Student ID:**

Question 1 (25):

a) For a steady laminar flow of a circular pipe, the discharge is given as below. Please, obtain the pressure drop due to energy loss. Express the relations between the energy loss and the related parameters.

$$Q = \frac{\gamma \pi D^4 J}{128 \mu}$$

b) Darcy-Weisbach equation is used to calculate fiction head loss in a pipe. Express the Darcy-Weisbach equation in terms of the discharge Q, pipe diameter D and length L.

c) Please draw the pump curve (Head versus Discharge) for any pump.

d) According to the Prandtl's Mixing Length Theory, how does the increase in the mixing length "L" affect the velocity field in a pipe cross-section?

e) What is the difference between $\tau_l = \mu \frac{du}{dy}$ and $\tau_t = \rho \varepsilon \frac{du}{dy}$? Explain.